二 不定积分的第一换元法（凑微分）

定理1 如果$u=g(x)$是可微函数，$f(u)$在$g(x)$的值域区间I上连续，则

$$\int f(g(x))g'(x)dx=\int f(g(x))d(g(x))$$

$$\overset{u=g(x)}{=}\int f(u)du$$

例 求 $\int \frac{1}{1+\cos x}dx.$

解 $\int \frac{1}{1+\cos x}dx = \int \frac{1-\cos x}{(1+\cos x)(1-\cos x)}dx$

$$= \int \frac{1-\cos x}{1-\cos^2 x}dx = \int \frac{1-\cos x}{\sin^2 x}dx$$

$$= \int \frac{1}{\sin^2 x}dx - \int \frac{1}{\sin^2 x}d(\sin x)$$

$$= -\cot x + \frac{1}{\sin x} + C.$$

总结(1)$\int \sin^{2m+1} x dx = \int \sin^{2m} x \sin x dx$

$$= -\int (1-\cos^2 x)^m d(\cos x)$$

(2)$\int \cos^{2m+1} x dx = \int (1-\sin^2 x)^m d(\sin x)$

(3)$\int \sin^{2m} x dx, \int \cos^{2m} x dx$ 利用倍角公式降幂

(4)$\int \sin \alpha x \cos \beta x dx, \int \sin \alpha x \sin \beta x dx, \int \cos \alpha x \cos \beta x dx$

利用积化和差公式

例　求　$\int \cos 3x \cos 2x dx.$

解　$\cos A \cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)],$

$$\cos 3x \cos 2x = \frac{1}{2}(\cos x + \cos 5x),$$

$$\int \cos 3x \cos 2x dx = \frac{1}{2}\int (\cos x + \cos 5x) dx$$

$$= \frac{1}{2}\sin x + \frac{1}{10}\sin 5x + C.$$

例 求$\int \sin^2 x \cdot \cos^5 x dx$.

解 $\int \sin^2 x \cdot \cos^5 x dx = \int \sin^2 x \cdot \cos^4 x d(\sin x)$

$$= \int \sin^2 x \cdot (1 - \sin^2 x)^2 d(\sin x)$$

$$= \int (\sin^2 x - 2\sin^4 x + \sin^6 x) d(\sin x)$$

$$= \frac{1}{3}\sin^3 x - \frac{2}{5}\sin^5 x + \frac{1}{7}\sin^7 x + C.$$

说明 当被积函数是三角函数相乘时，拆开奇次项去凑微分.

$$(1)\int \sin^{2m+1} x\cos^{2m} x dx = \int \sin^{2m} x\cos^{2m} x\sin x dx$$

$$= -\int (1-\cos^2 x)^m \cos^{2m} x d(\cos x)$$

$$(2)\int \cos^{2m+1} x\sin^{2m} x dx = \int \cos^{2m} x\sin^{2m} x\cos dx$$

$$= \int (1-\sin^2 x)^m \sin^{2m} x d(\sin x)$$

例 求 $\int \csc x dx.$

解（一）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{1}{2\sin\frac{x}{2}\cos\frac{x}{2}} dx$

$$= \int \frac{1}{\tan\frac{x}{2}\left(\cos\frac{x}{2}\right)^2} d\left(\frac{x}{2}\right) = \int \frac{1}{\tan\frac{x}{2}} d\left(\tan\frac{x}{2}\right)$$

$$= \ln\left|\tan\frac{x}{2}\right| + C = \ln\left|\csc x - \cot x\right| + C.$$

（使用了三角函数恒等变形）

解（二）$\int \csc x dx = \int \frac{1}{\sin x} dx = \int \frac{\sin x}{\sin^2 x} dx$

$$= -\int \frac{1}{1-\cos^2 x} d(\cos x) \qquad u = \cos x$$

$$= -\int \frac{1}{1-u^2} du = -\frac{1}{2}\int \left(\frac{1}{1-u} + \frac{1}{1+u} \right) du$$

$$= \frac{1}{2}\ln\left|\frac{1-u}{1+u}\right| + C = \frac{1}{2}\ln\left|\frac{1-\cos x}{1+\cos x}\right| + C.$$

类似地可推出 $\int \sec x dx = \ln\left|\sec x + \tan x\right| + C.$

例 设 $f'(\sin^2 x)=\cos^2 x$,求 $f(x)$.

解 令 $u=\sin^2 x \Rightarrow \cos^2 x=1-u$,

$$f'(u)=1-u,$$

$$f(u)=\int(1-u)du=u-\frac{1}{2}u^2+C,$$

$$f(x)=x-\frac{1}{2}x^2+C.$$

例 求 $\int \frac{1}{\sqrt{4-x^2}\arcsin\frac{x}{2}}dx.$

解 $\int \frac{1}{\sqrt{4-x^2}\arcsin\frac{x}{2}}dx = \int \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}\arcsin\frac{x}{2}}d\frac{x}{2}$

$$= \int \frac{1}{\arcsin\frac{x}{2}}d\left(\arcsin\frac{x}{2}\right) = \ln\left|\arcsin\frac{x}{2}\right| + C.$$

要熟记常见的微分式

$$xdx = \frac{1}{2}d(x^2), \qquad \frac{1}{ax+b}dx = \frac{1}{a}\ln(ax+b),(ax+b>0)$$

$$\frac{1}{\sqrt{x}}dx = 2d(\sqrt{x}), \quad \frac{1}{x^2}dx = -d(\frac{1}{x}), \qquad e^x d = d(e^x),$$

$$\sin xdx = -d(\cos x), \qquad \cos xdx = d(\sin x),$$ 等

例.求 $\int \sec x dx$.

解: $\int \sec x dx = \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx$

$$= \int \frac{1}{1-\sin^2 x} d\sin x = -\int \frac{1}{\sin^2 x - 1} d\sin x$$

$$= -\frac{1}{2}\ln\left|\frac{\sin x - 1}{\sin x + 1}\right| + C = \frac{1}{2}\ln\left|\frac{\sin x + 1}{\sin x - 1}\right| + C$$

$$= \frac{1}{2}\ln\frac{(\sin x + 1)^2}{\cos^2 x} + C = \ln\left|\frac{\sin x + 1}{\cos x}\right| + C$$

$$= \ln|\sec x + \tan x| + C.$$

同理: $\int \csc x dx = \ln|\csc x - \cot x| + C.$

例求 $\int \sec x dx.$

法二: $\int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx$

$$= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx = \int \frac{1}{\sec x + \tan x} d(\sec x + \tan x)$$

$$= \ln|\sec x + \tan x| + C.$$

例. $\int \frac{x+7}{x^2-4x-5}dx.$

解: $\frac{x+7}{x^2-4x-5} = \frac{x+7}{(x-5)(x+1)} = \frac{A}{x-5} + \frac{B}{x+1}$

$\Rightarrow \quad x+7 = A(x+1) + B(x-5),$

令$x=-1,\ \Rightarrow B=-1,$ 令$x=5,\ \Rightarrow A=2,$

$$\int \frac{x+7}{x^2-4x-5}dx = \int \left(\frac{2}{x-5} + \frac{-1}{x+1}\right)dx$$

$$= 2\ln|x-5| - \ln|x+1| + C$$

例. $\int \frac{x-2}{x^2+2x+3}dx.$

解: $\int \frac{x-2}{x^2+2x+3}dx = \int \frac{\frac{1}{2}(2x+2)-3}{x^2+2x+3}dx$

$$= \frac{1}{2}\int \frac{2x+2}{x^2+2x+3}dx - 3\int \frac{dx}{(x+1)^2+(\sqrt{2})^2}$$

$$= \frac{1}{2}\int \frac{1}{x^2+2x+3}d(x^2+2x+3) - 3\int \frac{d(x+1)}{(x+1)^2+(\sqrt{2})^2}$$

$$= \frac{1}{2}\ln(x^2+2x+3) - \frac{3}{\sqrt{2}}\arctan\frac{x+1}{\sqrt{2}} + C.$$

思考题

已知 $\frac{\cos x}{x}$ 是 $f(x)$ 的一个原函数，则

$$\int f(x)\frac{\cos x}{x}dx = \underline{\frac{1}{2}\left(\frac{\cos x}{x}\right)^2 + C}$$

解：$\because d\left(\frac{\cos x}{x}\right) = f(x)dx$

$$\therefore \int f(x)\frac{\cos x}{x}dx = \int \frac{\cos x}{x}d\left(\frac{\cos x}{x}\right) = \frac{1}{2}\left(\frac{\cos x}{x}\right)^2 + C$$

不定积分 $\int \frac{1}{\sqrt{x(1-x)}}dx =$ （ D

（A） $\frac{1}{2}\arcsin\sqrt{x}+C$ ；

（B） $\arcsin\sqrt{x}+C$ ；

（C） $2\arcsin(2x-1)+C$ ；

（D） $\arcsin(2x-1)+C$.

解： $\int \frac{1}{\sqrt{x(1-x)}}dx = \int \frac{1}{\sqrt{\frac{1}{4}-\left(x-\frac{1}{2}\right)^2}}dx = 2\int \frac{1}{\sqrt{1-(2x-1)^2}}dx$

$$= \int \frac{1}{\sqrt{1-(2x-1)^2}}d(2x-1) = \arcsin(2x-1)+C$$

不定积分的第二换元法

定理2　设$x = g(t)$是单调的可微函数，且$g'(t) \neq 0$，

$f(x)$连续，则$\int f(x)dx = \int f(g(t))g'(t)dt$

$$= \{F(t) + c\}_{t=g^{-1}(x)}$$

例. $\int \sqrt{1-x^2}dx$

解: 令$x=\sin t,\ -\frac{\pi}{2}\le t\le\frac{\pi}{2},$

$dx=\cos tdt,$

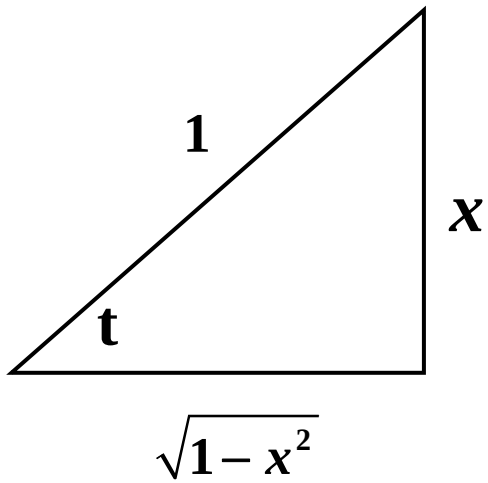
$$\int \sqrt{1-x^2}dx=\int \sqrt{1-(\sin t)^2}\cos tdt$$

$$=\int \cos t\cos tdt=\int \cos^2 tdt$$

$$=\int \frac{1+\cos 2t}{2}dt=\frac{1}{2}(t+\frac{1}{2}\sin 2t)+C$$

$$=\frac{1}{2}(t+\sin t\cos t)+C$$

$$=\frac{1}{2}(\arcsin x+x\sqrt{1-x^2})+C$$



例. $\int \sqrt{a^2 - x^2}dx \ (a > 0).$

解: 令$x = a\sin t$, $-\frac{\pi}{2} \le t \le \frac{\pi}{2}$,

则 $dx = a\cos t dt$,

$$\int \sqrt{a^2 - x^2}dx = \int \sqrt{a^2 - (a\sin t)^2}a\cos t dt$$

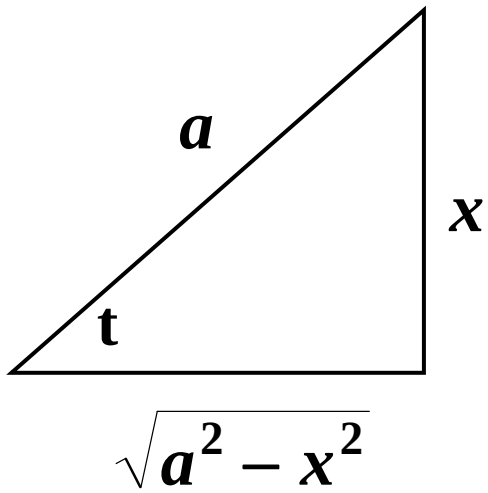
$$= \int a\cos t \cdot a\cos t dt \ = a^2 \int \cos^2 t dt$$

$$= a^2 \int \frac{1 + \cos 2t}{2} dt \ = \frac{a^2}{2}(t + \frac{1}{2}\sin 2t) + C$$

$$= \frac{a^2}{2}(t + \sin t\cos t) + C$$

$$= \frac{a^2}{2}(\arcsin\frac{x}{a} + \frac{x}{a}\frac{\sqrt{a^2 - x^2}}{a}) + C$$

$$= \frac{a^2}{2}\arcsin\frac{x}{a} + \frac{x}{2}\sqrt{a^2 - x^2} + C.$$



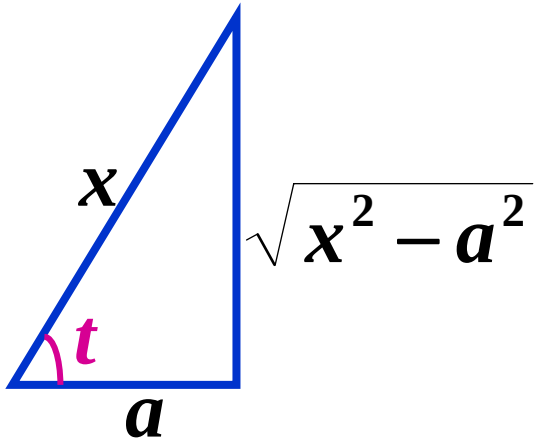
例 求 $\int \frac{1}{\sqrt{x^2-a^2}}dx \quad (a>0).$

解 令$x = a\sec t \quad dx = a\sec t\tan t dt \quad t \in \left(0, \frac{\pi}{2}\right)$

$$\int \frac{1}{\sqrt{x^2-a^2}}dx = \int \frac{a\sec t \cdot \tan t}{a\tan t}dt$$

$$= \int \sec t dt = \ln\left|\sec t + \tan t\right| + C$$

$$= \ln\left|\frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a}\right| + C.$$



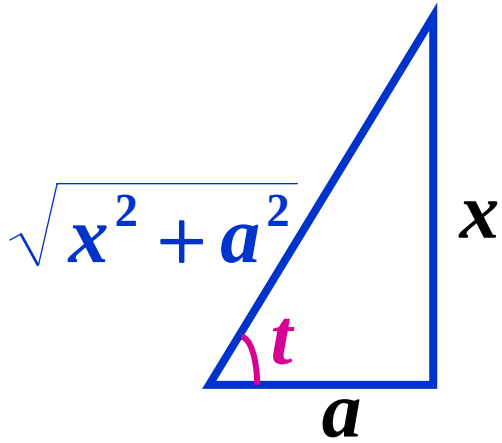
例 求 $\int \frac{1}{\sqrt{x^2+a^2}}dx \quad (a>0).$

解 令 $x = a\tan t \Rightarrow dx = a\sec^2 t dt \qquad t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$\int \frac{1}{\sqrt{x^2+a^2}}dx = \int \frac{1}{a\sec t}\cdot a\sec^2 t dt$$

$$= \int \sec t dt = \ln\left|\sec t + \tan t\right| + C$$

$$= \ln\left|\frac{x}{a} + \frac{\sqrt{x^2+a^2}}{a}\right| + C.$$



说明(1) 以上几例所使用的均为三角代换.

三角代换的目的是化掉根式.

一般规律如下：当被积函数中含有

(1) $\sqrt{a^2 - x^2}$ 可令 $x = a\sin t$;

(2) $\sqrt{a^2 + x^2}$ 可令 $x = a\tan t$;

(3) $\sqrt{x^2 - a^2}$ 可令 $x = a\sec t$.

例.求 $\displaystyle\int \frac{\sqrt{9x^2+6x-8}}{3x+1}dx.$

解: $\because \sqrt{9x^2+6x-8}=\sqrt{(3x+1)^2-3^2},$

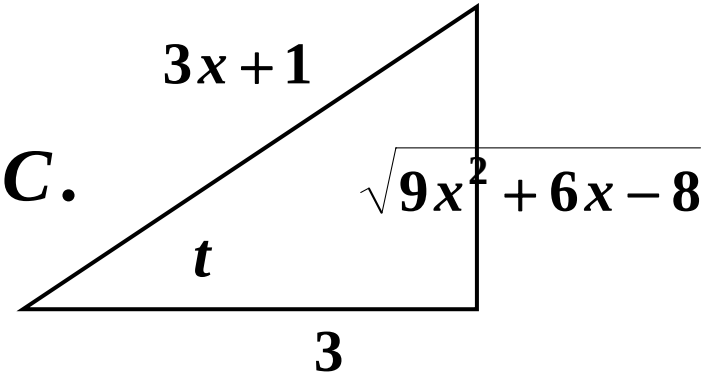
$\therefore$ 令 $3x+1=3\sec t,\ (0\le t<\dfrac{\pi}{2})$

则 $dx=\sec t\tan t dt,$

原积分 $=\displaystyle\int \frac{3\tan t}{3\sec t}\sec t\tan t dt=\int \tan^2 t dt$

$=\displaystyle\int(\sec^2 t-1)dt\ =\tan t-t+C$

$=\dfrac{\sqrt{9x^2+6x-8}}{3}-\arccos\dfrac{3}{3x+1}+C.$



答疑通知

- **1**月**18**日：（**1**）上午**9**：**00--11**：**30**；
- （**2**）下午**1:30--4:00**；
- **(3)** 晚上 **6**：**30--9**：**00**。
- 答疑地点：理学院楼**129**室（奉贤）

不定积分的第二换元法

定理2　设$x = g(t)$是单调的可微函数，且$g'(t) \neq 0$，

$f(x)$连续，则$\int f(x)dx = \int f(g(t))g'(t)dt$

$$= \{F(t) + c\}_{t=g^{-1}(x)}$$

说明(1) 三角代换.

三角代换的目的是化掉根式.

一般规律如下：当被积函数中含有

(1) $\sqrt{a^2 - x^2}$ 可令 $x = a\sin t$;

(2) $\sqrt{a^2 + x^2}$ 可令 $x = a\tan t$;

(3) $\sqrt{x^2 - a^2}$ 可令 $x = a\sec t$.

说明(2) 积分中为了化掉根式除采用三角代换外还可用双曲代换.

$$\because ch^2 t - sh^2 t = 1$$

$\therefore x = asht,\quad x = acht$ 也可以化掉根式

例 $\int \frac{1}{\sqrt{x^2+a^2}}dx$ 中, 令 $x = asht$ $\quad dx = achtdt$

$$\int \frac{1}{\sqrt{x^2+a^2}}dx = \int \frac{a\,\mathrm{ch}\,t}{a\,\mathrm{ch}\,t}dt = \int dt = t + C$$

$$= ar\,\mathrm{sh}\frac{x}{a} + C = \ln\left(\frac{x}{a} + \frac{\sqrt{x^2+a^2}}{a}\right) + C.$$

说明(3) 积分中为了化掉根式是否一定采用三角代换（或双曲代换）并不是绝对的，需根据被积函数的情况来定.

例 求 $\int \frac{x^5}{\sqrt{1+x^2}}dx$ （三角代换很繁琐）

解 令 $t=\sqrt{1+x^2} \Rightarrow x^2=t^2-1, xdx=tdt$,

$$\int \frac{x^5}{\sqrt{1+x^2}}dx = \int \frac{(t^2-1)^2}{t}tdt = \int (t^4-2t^2+1)dt$$

$$= \frac{1}{5}t^5 - \frac{2}{3}t^3 + t + C$$

$$= \frac{1}{15}(8-4x^2+3x^4)\sqrt{1+x^2} + C.$$

说明(4) 当分母的阶较高时, 可采用倒代换 $x=\frac{1}{t}$.

例 求 $\int \frac{1}{x(x^7+2)}dx$

解 令 $x=\frac{1}{t} \Rightarrow dx=-\frac{1}{t^2}dt,$

$$\int \frac{1}{x(x^7+2)}dx=\int \frac{t}{\left(\frac{1}{t}\right)^7+2}\cdot\left(-\frac{1}{t^2}\right)dt=-\int \frac{t^6}{1+2t^7}dt$$

$$=-\frac{1}{14}\ln|1+2t^7|+C=-\frac{1}{14}\ln|2+x^7|+\frac{1}{2}\ln|x|+C.$$

例. 求 $\int \frac{dx}{x(x^n+1)}(n \neq 0).$

解: 令 $x=\frac{1}{t},(t>0)$, (倒代换), 则 $dx=-\frac{1}{t^2}dt$,

$$\int \frac{dx}{x(x^n+1)}=\int \frac{-\frac{1}{t^2}dt}{\frac{1}{t}(\frac{1}{t^n}+1)}=-\int \frac{t^{n-1}}{1+t^n}dt$$

$$=-\frac{1}{n}\int \frac{1}{1+t^n}d(1+t^n)=-\frac{1}{n}\ln|1+t^n|+C$$

$$=-\frac{1}{n}\ln|1+\frac{1}{x^n}|+C.$$

例. 求 $\int \frac{dx}{x(x^n+1)} (n \neq 0).$

$$\int \frac{dx}{x(x^n+1)} = \int \left(\frac{1}{x} - \frac{x^{n-1}}{x^n+1} \right) dx$$

$$= \ln|x| - \frac{1}{n} \ln|1 + x^n| + C$$

说明(5) 当被积函数含有两种或两种以上的根式 $\sqrt[k]{x},\cdots,\sqrt[l]{x}$ 时，可采用令 $x=t^n$（其中 n 为各根指数的最小公倍数）

例 求 $\displaystyle\int\frac{1}{\sqrt{x}(1+\sqrt[3]{x})}dx.$

解 令 $x=t^6\Rightarrow dx=6t^5dt,$

$$\int\frac{1}{\sqrt{x}(1+\sqrt[3]{x})}dx=\int\frac{6t^5}{t^3(1+t^2)}dt=\int\frac{6t^2}{1+t^2}dt$$

$$= 6\int \frac{t^2+1-1}{1+t^2}dt$$

$$= 6\int \left(1-\frac{1}{1+t^2}\right)dt$$

$$= 6[t-\arctan t]+C$$

$$= 6[\sqrt[6]{x}-\arctan\sqrt[6]{x}]+C.$$

例求 $\int \frac{1}{\sqrt{1+e^{2x}}}dx.$

解: $\int \frac{1}{\sqrt{1+e^{2x}}}dx = \int \frac{1}{e^{x}\sqrt{1+e^{-2x}}}dx$

$$= \int \frac{e^{-x}}{\sqrt{1+e^{-2x}}}dx = \int \frac{-1}{\sqrt{1+(e^{-x})^{2}}}de^{-x}$$

$$= -\ln|e^{-x} + \sqrt{1+e^{-2x}}| + C.$$

例. $\int \frac{1}{1+\sqrt{e^x}}dx.$

解: 令 $\sqrt{e^x}=t,(t>0),$

则 $x=\ln t^2=2\ln t,\ dx=\frac{2}{t}dt,$

$$\int \frac{1}{1+\sqrt{e^x}}dx=\int \frac{1}{1+t}\times\frac{2}{t}dt$$

$$=2\int(\frac{1}{t}-\frac{1}{1+t})dt=2\ln t-2\ln(1+t)+C,$$

$$=2\ln\sqrt{e^x}-2\ln(1+\sqrt{e^x})+C.$$

基本积分表

(16) $\int \tan x dx = -\ln|\cos x| + C;$

(17) $\int \cot x dx = \ln|\sin x| + C;$

(18) $\int \sec x dx = \ln|\sec x + \tan x| + C;$

(19) $\int \csc x dx = \ln|\csc x - \cot x| + C;$

(20) $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C;$

(21) $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C;$

(22) $$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \ln \left| \frac{a + x}{a - x} \right| + C;$$

(23) $$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C;$$

(24) $$\int \frac{1}{\sqrt{x^2 \pm a^2}} dx = \ln \left| x + \sqrt{x^2 \pm a^2} \right| + C.$$

三、小结

两类积分换元法：

$$\begin{cases} \text{（一）凑微分} \\ \text{（二）三角代换、倒代换、根式代换} \end{cases}$$

四　分部积分法

定理．设函数$f(x),g(x)$具有连续的导数 $f'(x),g'(x)$,

则$$\int f(x)dg(x)=f(x)g(x)-\int g(x)df(x)$$

即$$\int f(x)g'(x)dx=f(x)g(x)-\int g(x)f'(x)dx$$

例1　求积分 $\int x\cos x dx$.

解（一）　令 $u=\cos x$,　　$xdx=\frac{1}{2}dx^2=dv$

$$\int x\cos x dx=\frac{x^2}{2}\cos x+\int\frac{x^2}{2}\sin x dx$$

显然，u,v' 选择不当，积分更难进行.

解（二）　令 $u=x$,　$\cos x dx=d\sin x=dv$

$$\int x\cos x dx=\int x d\sin x=x\sin x-\int\sin x dx$$

$$=x\sin x+\cos x+C.$$

例 求积分 $\int xe^x dx.$

总结 若被积函数是幂函数和正(余)弦函数或幂函数和指数函数的乘积，就考虑设幂函数为**u**, 使其降幂一次(假定幂指数是正整数)

例 求积分 $\int x^2 e^x dx.$

解 $u = x^2,\qquad e^x dx = de^x = dv,$

$$\int x^2 e^x dx = x^2 e^x - 2\int x e^x dx$$

（再次使用分部积分法）$u = x,\quad e^x dx = dv$

$$= x^2 e^x - 2(xe^x - e^x) + C.$$

例　求积分 $\int x\arctan x dx$.

解　令 $u=\arctan x$,　$xdx=d\frac{x^2}{2}=dv$

$$\int x\arctan x dx=\frac{x^2}{2}\arctan x-\int\frac{x^2}{2}d(\arctan x)$$

$$=\frac{x^2}{2}\arctan x-\int\frac{x^2}{2}\cdot\frac{1}{1+x^2}dx$$

$$=\frac{x^2}{2}\arctan x-\int\frac{1}{2}\cdot(1-\frac{1}{1+x^2})dx$$

$$=\frac{x^2}{2}\arctan x-\frac{1}{2}(x-\arctan x)+C.$$

例　求积分 $\int\arctan xdx$　$\int\arcsin xdx$

例　求积分 $\int x^3 \ln x dx.$

解　$u = \ln x,\quad x^3 dx = d\frac{x^4}{4} = dv,$

$$\int x^3 \ln x dx = \frac{1}{4}x^4 \ln x - \frac{1}{4}\int x^3 dx$$

$$= \frac{1}{4}x^4 \ln x - \frac{1}{16}x^4 + C.$$

总结　若被积函数是幂函数和对数函数或幂函数和反三角函数的乘积，就考虑设对数函数或反三角函数为 u.

例　求积分 $\int \frac{x\arctan x}{\sqrt{1+x^2}}dx.$

解　$\because \quad \left(\sqrt{1+x^2}\right)' = \frac{x}{\sqrt{1+x^2}},$

$$\therefore \quad \int \frac{x\arctan x}{\sqrt{1+x^2}}dx = \int \arctan x d\sqrt{1+x^2}$$

$$= \sqrt{1+x^2}\arctan x - \int \sqrt{1+x^2}d(\arctan x)$$

$$= \sqrt{1+x^2}\arctan x - \int \sqrt{1+x^2}\cdot\frac{1}{1+x^2}dx$$

$$= \sqrt{1+x^2}\arctan x - \boxed{\int \frac{1}{\sqrt{1+x^2}}dx}$$ 令 $x = \tan t$

$$\int \frac{1}{\sqrt{1+x^2}}dx = \int \frac{1}{\sqrt{1+\tan^2 t}}\sec^2 t dt = \int \sec t dt$$

$$= \ln(\sec t + \tan t) + C = \ln(x + \sqrt{1+x^2}) + C$$

$$\therefore \int \frac{x\arctan x}{\sqrt{1+x^2}}dx$$

$$= \sqrt{1+x^2}\arctan x - \ln(x + \sqrt{1+x^2}) + C.$$

例已知 $f(x)$ 的一个原函数是 e^{-x^2}，求 $\int xf'(x)dx$.

解 $\int xf'(x)dx = \int xdf(x) = xf(x) - \int f(x)dx,$

$\because \left(\int f(x)dx\right)' = f(x), \therefore \int f(x)dx = e^{-x^2} + C,$

两边同时对 x 求导, 得 $f(x) = -2xe^{-x^2},$

$$\therefore \int xf'(x)dx = xf(x) - \int f(x)dx$$

$$= -2x^2e^{-x^2} - e^{-x^2} + C.$$

例　求积分 $\int e^x \sin x dx$.

解 $\boxed{\int e^x \sin x dx} = \int \sin x de^x$

$$= e^x \sin x - \int e^x d(\sin x)$$

$$= e^x \sin x - \int e^x \cos x dx$$

$$= e^x \sin x - \int \cos x de^x$$

$$= e^x \sin x - (e^x \cos x - \int e^x d\cos x)$$

$$= e^x (\sin x - \cos x) - \boxed{\int e^x \sin x dx}$$

注意循环形式

$$\therefore \int e^x \sin x dx = \frac{e^x}{2}(\sin x - \cos x) + C.$$

在接连几运用分部积分公式时，

应注意什么？

前后几次所选的 **u (x)** 应为同类型函数**.**

例 $$\boxed{\int \sec^3 x dx} = \int \sec x \sec^2 x dx$$

$$= \int \sec x d \tan x$$

$$= \sec x \tan x - \int \tan^2 x \sec x dx$$

$$= \sec x \tan x - \int (\sec^2 x - 1) \sec x dx$$

$$= \sec x \tan x - \boxed{\int \sec^3 x dx} + \int \sec x dx$$

例　求积分 $\int \sin(\ln x)dx.$

解 $\boxed{\int \sin(\ln x)dx} = x\sin(\ln x) - \int x d[\sin(\ln x)]$

$$= x\sin(\ln x) - \int x\cos(\ln x)\cdot\frac{1}{x}dx$$

$$= x\sin(\ln x) - x\cos(\ln x) + \int x d[\cos(\ln x)]$$

$$= x[\sin(\ln x) - \cos(\ln x)] - \boxed{\int \sin(\ln x)dx}$$

$$\therefore \int \sin(\ln x)dx = \frac{x}{2}[\sin(\ln x) - \cos(\ln x)] + C.$$

例. 建立 $I_n=\int\frac{dx}{(x^2+a^2)^n}$ 的递推公式.

解: $I_n=\int\frac{dx}{(x^2+a^2)^n}=\frac{x}{(x^2+a^2)^n}-\int xd\frac{1}{(x^2+a^2)^n}$

$$=\frac{x}{(x^2+a^2)^n}+2n\int\frac{x^2+a^2-a^2}{(x^2+a^2)^{n+1}}dx$$

$$=\frac{x}{(x^2+a^2)^n}+2n\int\frac{1}{(x^2+a^2)^n}dx-2na^2\int\frac{1}{(x^2+a^2)^{n+1}}dx$$

$$=\frac{x}{(x^2+a^2)^n}+2nI_n-2na^2I_{n+1},$$

$$I_{n+1}=\frac{x}{2na^2(x^2+a^2)^n}+\frac{2n-1}{2a^2n}I_n.$$

于是有: $I_n = \dfrac{x}{2(n-1)a^2(x^2+a^2)^{n-1}} + \dfrac{2n-3}{2a^2(n-1)} I_{n-1},$

和 $I_{n-1} = \dfrac{x}{2(n-2)a^2(x^2+a^2)^{n-2}} + \dfrac{2n-5}{2a^2(n-2)} I_{n-2}, \cdots$

$$I_2 = \frac{x}{2a^2(x^2+a^2)} + \frac{1}{2a^2} I_1$$

而 $I_1 = \int \dfrac{1}{x^2+a^2} dx = \dfrac{1}{a} \arctan \dfrac{x}{a} + C,$

再由递推公式可依次求 出: $I_2, I_3, \cdots$

指三幂对反